SIGMACAMP · ELECTROCHEMISTRY LAB · SESSION 4

# Water Electrolysis

Experiment Packet

**Audience:** instructors and teaching assistants

**Purpose:** Bench instructions for instructors and TAs

**Session hook:** Split water and see a 2:1 ratio

**Length:** 90-minute lab block (~20 min concepts + ~70 min hands-on)

### This packet includes



Generated from the Electrochemistry workshop curriculum. Prefer the latest PDF from the course repository if procedures change mid-week.

SESSION OVERVIEW & PLAN

# Session 4 — Water Electrolysis and Gas Collection

**Hook:** Split water, collect gases, and see a 2:1 stoichiometric ratio.

## Learning outcomes

- Build a gas collection apparatus by water displacement
- Run water electrolysis safely with a non-chloride electrolyte
- Measure $H_2$ and $O_2$ volumes and compare to 2:1 ratio
- Connect electrolysis to hydrogen as an energy storage concept
- **Start** electrolytic rust-removal cells for Session 5 reveal

## Session plan (90 min)

| Time | Activity | Reference |
| --- | --- | --- |
| 0–10 min | Hook: inverted water-filled tubes — which gas appears faster? | lecture.md |
| 10–25 min | Water splitting and stoichiometry | lecture.md |
| 25–40 min | Build gas collection setup | experiment.md — Part A |
| 40–60 min | Run electrolysis; record volumes over time | experiment.md — Part B |
| 60–70 min | Discuss ratio; optional instructor $H_2$ pop test | experiment.md — Part C |
| 70–80 min | Brief energy storage discussion | lecture.md |
| 80–90 min | **Start derusting cells** for Session 5 | experiment.md — Part E |

## Key reactions

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Overall: 2H₂O(l) → 2H₂(g) + O₂(g)

Cathode (−): 2H₂O + 2e⁻ → H₂ + 2OH⁻
Anode (+):    2H₂O → O₂ + 4H⁺ + 4e⁻   (acidic media)
         or 4OH⁻ → O₂ + 2H₂O + 4e⁻   (basic media)
```

## Status

- [ ] Graphite electrodes sourced
- [ ] Gas collection apparatus pilot-tested
- [ ] **No saltwater** — baking soda or $Na_2SO_4$ only
- [ ] Pop test protocol reviewed
- [ ] Rusty nails + washing soda ready for Part E

PREPARATION

# Session 4 — Preparation

## Before class (instructor)

- [ ] **Pilot full setup** — confirm 2:1 ratio visible within session time
- [ ] Pre-cut graphite electrodes to useful lengths
- [ ] Mix electrolyte batch: water + baking soda (label "NO SALT")
- [ ] Test tubes: practice inversion technique — no trapped air
- [ ] Review pop test safety; decide if demo will run (weather, room rules)
- [ ] Optional: test solar panel output with same cell
- [ ] Stage rusty nails, washing soda, sacrificial anodes, and spare jars for **Part E** (end of class)
- [ ] Confirm overnight power policy for derusting cells (prep room vs. disconnect)

## Day-of setup

- [ ] Pre-fill tub at each station (or central fill station)
- [ ] Pre-fill inverted tubes underwater as demo for students
- [ ] Large sign: **"No salt in electrolysis"**
- [ ] Separate instructor area for pop test marked off
- [ ] Part E station: rusty nails, washing soda, steel/graphite anodes, labels, camera

## Common failure modes (from pilot)

| Issue | Prevention |
| --- | --- |
| Air in tubes | Demo inversion step slowly |
| Weak bubble rate | Fresh baking soda; clean graphite |
| Students add salt "to speed up" | Pre-brief; labeled solutions |

## Open questions / notes

- Electrode material used:
- Best tube type for measurement:
- Pop test yes/no for this venue:

MATERIALS CHECKLIST

# Session 4 — Materials

## Per group

- [ ] Clear plastic tub or container
- [ ] Water (~500 mL per tub)
- [ ] Baking soda: **1–2 tsp per 500 mL** (~0.1–0.2 M) **or** sodium sulfate ~0.1 M (~7 g / 500 mL)
- [ ] Graphite electrodes (thick pencil leads or carbon rods)
- [ ] 9 V battery or DC power supply
- [ ] Alligator clips
- [ ] Two small test tubes, vials, syringes, or narrow plastic bottles
- [ ] Ruler (volume estimation)
- [ ] Goggles

## Optional

- [ ] pH indicator
- [ ] Small solar panel
- [ ] Graduated cylinders for calibration

## Instructor-only (pop test)

- [ ] Small flame source (long lighter)
- [ ] Separate tiny collection vessel
- [ ] Fire extinguisher accessible

## Part E — Start rust removal (end of class)

- [ ] Rusty iron nails or hardware (2+ per group)
- [ ] Washing soda: **1 tbsp per ~250 mL** warm water (preferred) **or** baking soda same measure (NOT NaCl)
- [ ] Sacrificial steel strips or graphite rods
- [ ] Small jars or cups
- [ ] Labels and camera for before photos

## Explicitly excluded

- [ ] ~~Table salt (NaCl)~~ — **do not use**

## Pilot notes

| Parameter | Value from pilot |
| --- | --- |
| Electrolyte | Baking soda / $Na_2SO_4$ |

| Parameter | Value from pilot |
| --- | --- |
| Time to visible gas difference | ___ min |
| Typical $V\_H_2/V\_O_2$ | ___ |

EXPERIMENT PROCEDURE

# Session 4 — Experiment: Water Electrolysis

> **Instructors:** gas ID, ratio systematics, Part E polarity — instructor.md.

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## Electrolyte recipes — concentrations

**Never use table salt (NaCl)** — chloride can produce toxic chlorine gas at the anode.

### Option A — Baking soda (recommended, easy)

| Batch | Baking soda ($NaHCO_3$) | Water |
|---|---|---|
| ~500 mL (one tub) | **1–2 tsp** (~5–10 g) | ~500 mL |
| ~1 L | **2–4 tsp** (~10–20 g) | ~1 L |

Stir until dissolved. Label **"baking soda electrolyte — NO SALT"**.

Approximate concentration: **~0.1–0.2 M $NaHCO_3$** (kitchen measuring is fine; exact molarity is not critical).

**Water:** tap water is fine — the baking soda (or $Na_2SO_4$) provides the ions you need. Distilled water is unnecessary here.

### Option B — Sodium sulfate (optional)

| Batch | $Na_2SO_4$ (anhydrous) | Water |
|---|---|---|
| 500 mL | **~7 g** | 500 mL |

≈ **0.1 M $Na_2SO_4$**. Clear, effective supporting electrolyte.

### Volume per station

Fill the tub deep enough to fully cover inverted tube mouths (~3–5 cm liquid depth typical). Mix **fresh** electrolyte if someone contaminated a tub with salt.

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## Part A — Build gas collection setup (25–40 min)

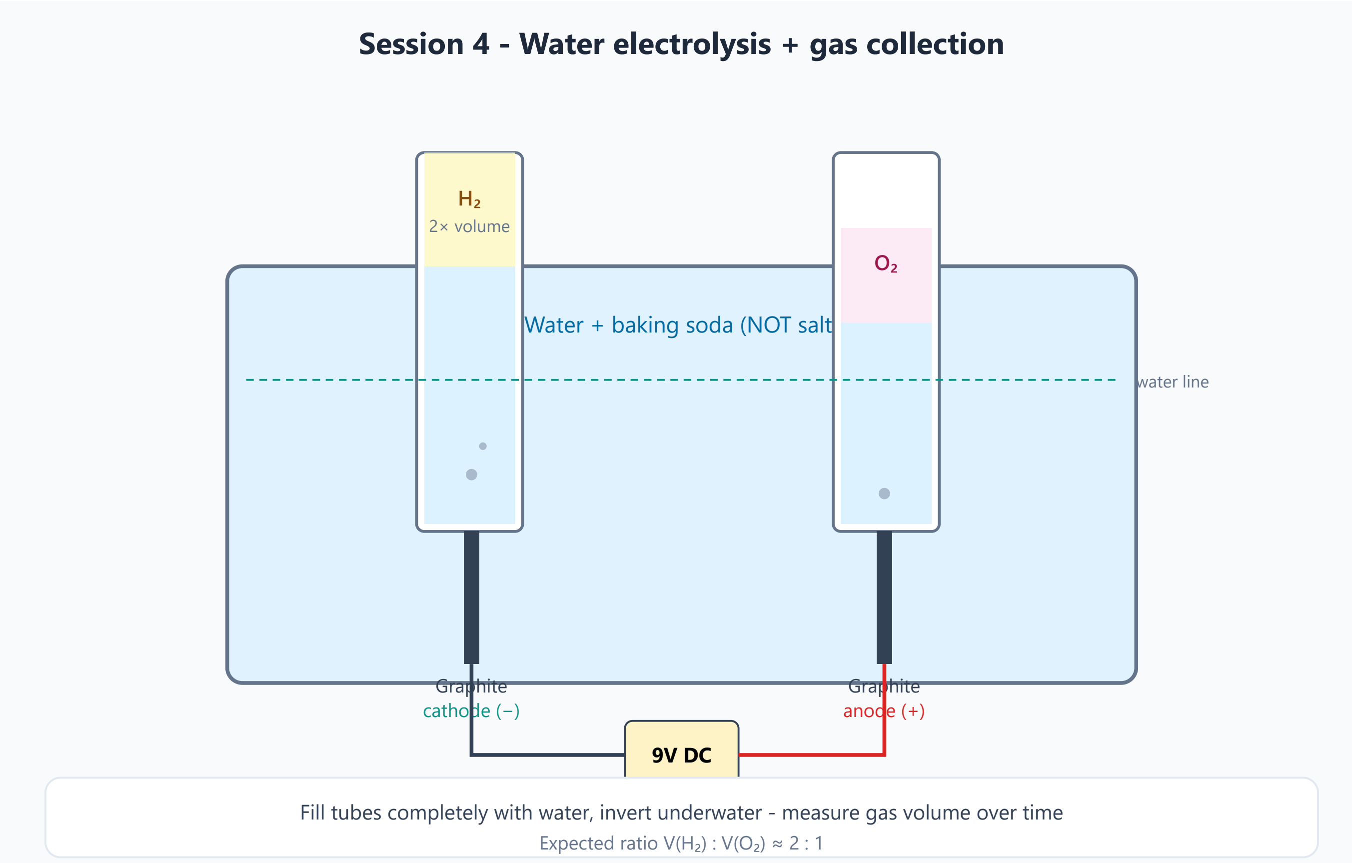


*Figure 1 — Inverted water-filled tubes collect $H_2$ (2× volume) and $O_2$. Use baking soda or sodium sulfate — not salt.*

### Materials

- Clear tub + electrolyte (Option A or B)
- Two small test tubes, vials, or narrow bottles (same size preferred)
- Two graphite electrodes (pencil leads ~2–3 mm or carbon rods)
- 9 V battery + alligator clips
- Ruler or graduated markings for gas column height
- Goggles

### Build steps

1. Fill the tub with electrolyte.
2. Fill tube 1 **completely** with the same electrolyte — no air bubble at the closed end.
3. Cover the mouth with a finger or card; invert; lower into the tub; release underwater. Tube stays full of liquid.
4. Repeat for tube 2.
5. Slide a graphite electrode up into each tube (or position electrodes so bubbles rise into each tube).
6. Connect:

7. **Cathode (−)** / black → electrode in tube that will collect **$H_2$** (will fill faster)
8. **Anode (+)** / red → other electrode (**$O_2$**)
9. Keep battery clips **dry** (not submerged).
10. When current flows, the **faster-filling** tube is hydrogen (cathode).

#### Pre-run checklist

- [ ] Both tubes liquid-filled with **no air gap** at start
- [ ] Electrodes secure; battery clips dry
- [ ] Electrolyte is baking soda or $Na_2SO_4$ — **not salt**
- [ ] Goggles on
- [ ] Tubes labeled $H_2$ / $O_2$ after first bubbles appear

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### Part B — Run and measure (40–65 min)

1. Start the timer when bubbles are clearly visible at both electrodes.
2. Every **2–3 min**, measure gas volume in each tube:
3. Prefer graduated tubes (mL), **or**
4. Measure gas column height (cm) with a ruler (ratio of heights ≈ ratio of volumes if tubes match).
5. Continue until a clear difference appears or ~15–20 min have elapsed.

#### Gas volume log

| Time (min) | $V\_H_2$ (mL or cm) | $V\_O_2$ (mL or cm) | Ratio $V\_H_2/V\_O_2$ |
|---|---|---|---|
| 0 | 0 | 0 | — |
| 3 | | | |
| 6 | | | |
| 9 | | | |
| 12 | | | |
| 15 | | | |
| 20 | | | |

#### Expected result

$V\_H_2$ / $V\_O_2$ ≈ **2** (ideal). Classroom range **1.6–2.2** is a success — discuss solubility and collection errors.

#### Observations

- Bubble rate at each electrode
- Cloudiness or electrode crumbling
- Temperature change (optional)

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## Part C — Instructor-only pop test (65–75 min)

### Optional demo — not for students

1. Collect a **tiny** amount of $H_2$ in a separate small tube.
2. Move away from the main tub and faces.
3. Brief flame → soft "pop."
4. Never ignite sealed or mixed $H_2/O_2$ containers.
5. Skip if the venue forbids open flame.

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## Part D — Extensions (if time)

- [ ] Add universal indicator — look for acid/base zones near electrodes
- [ ] Power with a small solar panel instead of a battery
- [ ] Compare baking soda vs. 0.1 M sodium sulfate side by side

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## Part E — Start electrolytic rust removal (80–90 min)

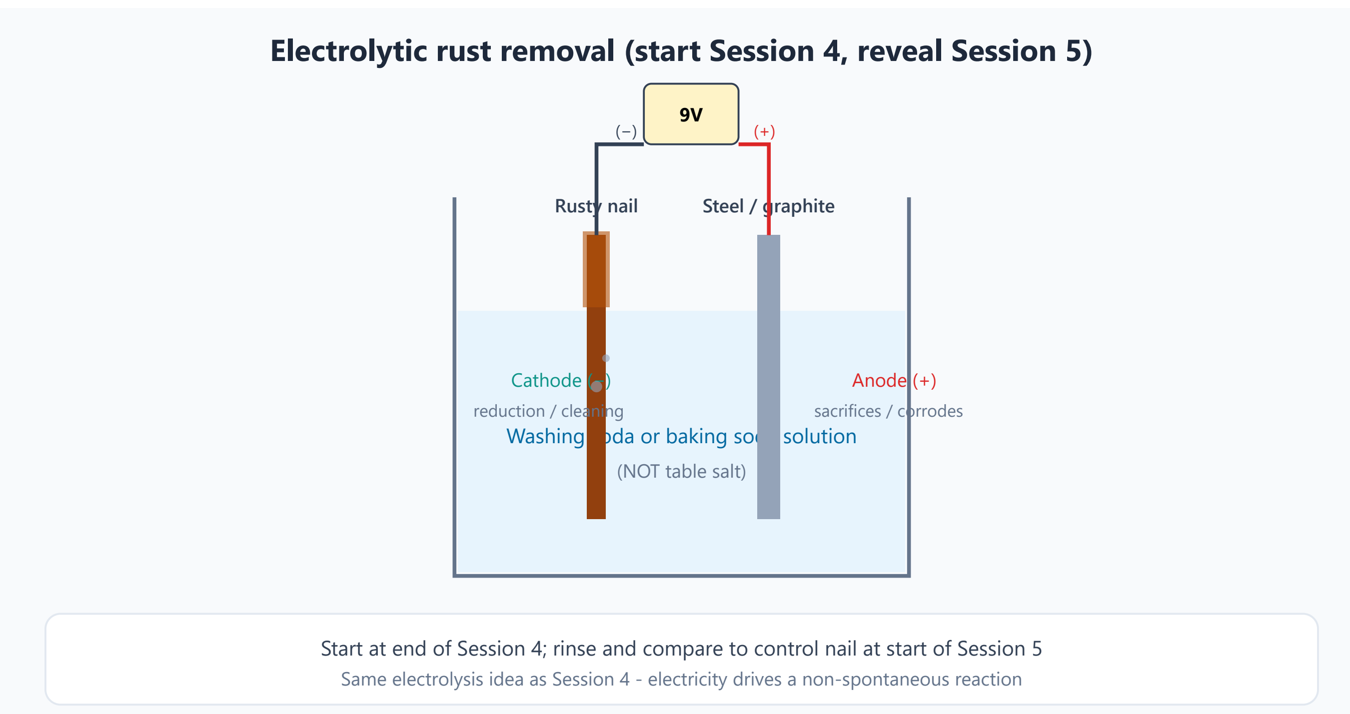


*Figure 2 — Start cells now; reveal cleaned nails at opening of Session 5.*

### Derusting electrolyte

| Ingredient | Amount |
|---|---|
| **Washing soda** (sodium carbonate, $Na_2CO_3$) preferred | **1 tbsp (~15 g) per ~250 mL** warm water |
| **or baking soda** | **1 tbsp per ~250 mL** warm water |

≈ **0.5–0.6 M** $Na_2CO_3$ if using washing soda at that kitchen measure — exact molarity is not critical. **Do NOT use NaCl.**

#### Setup steps

1. Dissolve washing soda (or baking soda) in warm water; pour into a small jar (~100–250 mL).
2. Clip **rusty iron** nail/hardware to battery **cathode (−)**.
3. Clip **sacrificial steel strip or graphite** to battery **anode (+)**.
4. Immerse both; keep them from touching; confirm gentle bubbling.
5. Label group name; photograph the nail **before**.
6. Place a **control** rusty nail in plain water (no battery).
7. Leave powered overnight per instructor / venue rules.

#### End-of-class checklist

- [ ] All derusting cells connected and bubbling (or current confirmed)
- [ ] Control nails in plain water labeled
- [ ] Before photos taken
- [ ] Overnight power policy confirmed (prep room preferred)

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## Troubleshooting

| Problem | Cause | Fix |
|---|---|---|
| No bubbles | Poor contact / weak electrolyte | Check clips; add more baking soda |
| One side only | Single electrode connected | Check both leads |
| Ratio far from 2:1 | Air in tube at start; leaks | Rebuild tubes with no air gap |
| Chlorine smell | **Salt used** | Stop; dump; remake baking soda bath |
| Tubes won't stay full | Air introduced on invert | Practice underwater release |

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## Safety checklist

- [ ] No NaCl electrolyte at this station
- [ ] No flames near the collection setup (except instructor pop test)
- [ ] Small gas volumes only
- [ ] Pop test instructor-only, at a safe distance

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## Experiment status

- [ ] Electrolyte recipe posted (baking soda amounts)
- [ ] Tube sizing / volume method chosen
- [ ] $H_2$ vs. $O_2$ side identified in pilot (cathode = $H_2$)

- [ ] Typical ratio from pilot: ___
- [ ] Rust-removal jars + washing soda staged for Part E

SHARED SAFETY NOTES

# Syllabus Safety Notes

Apply these across all sessions. Session-specific hazards are listed in each session folder.

## General rules

1. **No eating experimental materials.** Fruits, potatoes, and liquids used in experiments are no longer food.
2. **Avoid short circuits.** Batteries can heat up if directly shorted.
3. **Clean surfaces matter.** Electroplating requires clean metal surfaces.
4. **Metal salt waste should be collected.** Copper and silver solutions should not be casually poured down the drain.
5. **Water choice:** tap water is fine for most baths; use **distilled/deionized water for 0.01 M $AgNO_3$** (Session 3) so chloride does not precipitate silver.

## Session-specific hazards

| Session | Key hazard | Mitigation |
|---|---|---|
| 1 | Sharp nails/wires | Supervise handling; wash hands after metals |
| 2 | Copper solutions | Goggles; no ingestion; collect waste |
| 3 | Silver nitrate | Gloves + goggles; stains skin/clothing; small volumes |
| 4 | Gas evolution | No saltwater; no flames near setup; instructor-only pop test |
| 5 | Corrosion demos + electrolytic derusting | Use washing soda or baking soda only — NOT NaCl; disconnect power when not supervised |

## Hydrogen pop test (Session 4, instructor-only)

- Use a **tiny** volume of hydrogen only
- Perform away from the electrolysis setup
- **Never** ignite a sealed container or a mixed $H_2/O_2$ container